

Ex 7.1 (Symmetric matrices)

(H.W 35)

Only square matrix can be symmetric.

for a matrix A to be orthogonal $A^T A = I$.

This make $P^{-1} = P^T$.

→ for diagonalization. Find characteristic eq. extract Eigen values. use those eigen values to get eigenspaces.

diff.

→ all eigen values will give orthogonal space to each other.

But if we get two spaces from a same eigen value. It won't be orthogonal, Apply Gram Smidth to make it orthogonal.

→ Now normalize all the basis.

$$P = [u_1 \ u_2 \ \dots \ u_n]$$

$$D = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

$$A = P D P^{-1} = P D P^T \quad \therefore P^{-1} = P^T$$

$$\Rightarrow P^T A = D P^T$$

Ex 7.2 (Quadratic forms)

A quadratic form is a fn. Q whose values at vector x can be computed as $Q(x) = x^T A x$.

example $\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$ It should be symmetric matrix $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

Now substitute any vector x . diagonal value represent cross prod.

$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1^2 + 4x_1x_2 + 3x_2^2$$

$$2) 3x_1^2 + 2x_2^2 - 5x_3^2 - 6x_1x_2 + 8x_1x_3 - 4x_2x_3$$

$$\begin{bmatrix} 3 & -3 & 4 \\ -3 & 2 & -2 \\ 4 & -2 & -5 \end{bmatrix}$$

$$x = Py$$

(Removing cross product terms)

Change of Variable: That transforms quadratic form with cross product term with no cross product term.

for matrix $A = \begin{bmatrix} 1 & 5 \\ 5 & 1 \end{bmatrix}$. Find eigen values of A

which are 6 & -4. find eigen spaces for both & normalize it. Then make form $A = PDP^{-1}$.

same as in 7.1

where P is orthonormal basis & D is diagonal matrix with eigen values.

Now, Desired change of variable $x = Py$

$$\text{new form } x^T A x = (Py)^T A (Py) = y^T P^T A P y = y^T D y$$

A is symmetric so it is orthogonal matrix
 $P^{-1} = P^T$ & $D = P^T A P$

$$\text{As in this case } D = \begin{bmatrix} 6 & 0 \\ 0 & -4 \end{bmatrix}, \text{ so } \boxed{6y_1^2 - 4y_2^2 = y^T D y}$$

Classification of quadratic form.

(H.W 38)

- 1) Positive definite: All λ 's +ive.
- 2) Negative definite: " " -ive.
- 3) Indefinite: λ 's are both +ive & -ive.
- 4) positive semidefinite: λ 's are 0 or +ive.
- 5) negative semidefinite: " " " -ive.

Ex 7.3

find change of var. $x = Py$ that transform $x^T A x$ to $y^T D y$.

eg) $5x_1^2 + 6x_2^2 + 7x_3^2 + 4x_1x_2 - 4x_2x_3 = 9y_1^2 + 6y_2^2 + 3y_3^2$

Matrix $A = \begin{bmatrix} 5 & 2 & 0 \\ 2 & 6 & -2 \\ 0 & -2 & 7 \end{bmatrix}$, R.H.S implies $\lambda = 9, 6, 3$

Calculate eigen vectors normalized for each eigen value.

Those vectors will form P . $P = [v_1 \ v_2 \ v_3]$.

for max & min

In basic structure the constraint will always be $\|x\|=1$ ($x^T x = 1$)

- Greatest eigen value is the maximum value. (normalise it)
- Eigen vector of greatest eigen value is maximum vector (obv.)
- If constraint $x^T u_1 = 0$ is added i.e. It must be orthogonal to first (λ_1) so second eigen value will be max.
- Smallest eigen value will be minimum.

(H.W 38)

Problem 3 find max/min of $z = 4xy$ w.r.t $9x^2 + 4y^2 = 36 \Rightarrow \frac{x^2}{4} + \frac{y^2}{9} = 1$
 $x_1 = \frac{x}{2}, y_1 = \frac{y}{3}$

we only know on way. By $\|x\|=1$. So we convert it into our requirement.

$$z = 4xy = 4(2x_1)(3y_1) = 24x_1y_1 \quad : \quad x_1 = \frac{x}{2}, y_1 = \frac{y}{3}$$

Solve as mentioned above & at the end replace with actual terms.

Ex 7.4 SVD's

Eigen vectors of $A^T A$ gives V matrix.
" " $A A^T$ " U matrix

$$\sigma = \sqrt{\lambda}$$
$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_n \end{bmatrix}$$

$$A = U \Sigma V^T$$

If A is horizontal or square use $A^T A$.

Solve it for eigen values λ_i vectors (unit) normalized
This will give matrix Σ λ_i V .

for matrix U : $u_1 = \frac{1}{\sigma_1} A$

for last vector $u_1 \cdot u_3 = 0$, $u_2 \cdot u_3 = 0$

u_3 is arbitrary
vector like $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$

These eq. will give u_3 after solving.

Ex 7.5, Application to Image processing

Convert matrix of obs. to mean deviation form & construct the sample covariance matrix.

$$1, \begin{bmatrix} 19 & 22 & 6 & 3 & 2 & 20 \\ 12 & 6 & 9 & 15 & 13 & 5 \end{bmatrix} = X$$

$$\text{Sample Mean} = M = \frac{1}{\text{no. of col.}} \left[\begin{array}{c} \text{all} \\ \text{add of col.} \\ \text{of matrix} \end{array} \right]$$

$$M = \frac{1}{6} \begin{bmatrix} 72 \\ 60 \end{bmatrix} = \begin{bmatrix} 12 \\ 10 \end{bmatrix}$$

Mean deviation = subtract M
from B from each col of X

$$= \begin{bmatrix} 7 & 10 & -6 & -9 & -10 & 8 \\ 2 & -4 & -1 & 5 & 3 & -5 \end{bmatrix}$$

$$\text{Sample matrix (S)} = \frac{1}{6-1} BB^T = \frac{1}{5} \begin{bmatrix} 480 & -135 \\ -135 & 80 \end{bmatrix} = \begin{bmatrix} 96 & -27 \\ -27 & 16 \end{bmatrix}$$

no. of col.

Principal components: of data are the unit eigen vectors of S matrix. Compute eigen values of S (in descending order).

$$\lambda_1 = 95.2, \lambda_2 = 6.7, v_1 = \begin{bmatrix} -2.93348 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} .340892 \\ 1 \end{bmatrix}$$

Normalize these vectors for principal components.

$$u_1 = \begin{bmatrix} .946 \\ -.322 \end{bmatrix} \text{ for } \lambda_1 = 95.2 \quad \text{and} \quad u_2 = \begin{bmatrix} .322 \\ .946 \end{bmatrix} \text{ for } \lambda_2 = 6.79$$

x_1 & x_2 denote variables for 2D data. Find new variable y_1 of form $y_1 = c_1 x_1 + c_2 x_2$ with $c_1^2 + c_2^2 = 1$ such that y_1 has max possible variance over given data.

Since unit eigen vector corresponding to $\lambda_1 = 95.2$ is $u_1 = \begin{bmatrix} .946 \\ -.322 \end{bmatrix}$

one choice of new variable is $y_1 = .946 x_1 - .322 x_2$.

fraction of total variances. $\frac{\lambda_1}{\text{tr}(S)} = \frac{95.2}{(86+16)} = .933$, so 93.3% of

variance of data is explained by y_1 .